ClipMind AI – Milestone 3

Key Moments Detection & Analytics Dashboard

1. Topic Segmentation

- **Chunking Strategy**: Transcripts are divided into segments capped at 900 characters, ensuring no source segment is split.
- **Similarity Comparison**:
    - Primary: Normalized all-MiniLM-L6-v2 embeddings.
    - Fallback: Token-overlap Jaccard similarity when embeddings are unavailable.
    - similarity = intersection of meaningful words / union of meaningful words

  The fallback removes common stop words such as: the, and, to, is, of, in
- **Boundary Threshold**: A similarity score below 0.22 triggers a new topic boundary.
- **Topic Metadata**: Each topic stores:
    - Timestamp range
    - First-sentence title (auto-generated)
    - Full transcript context

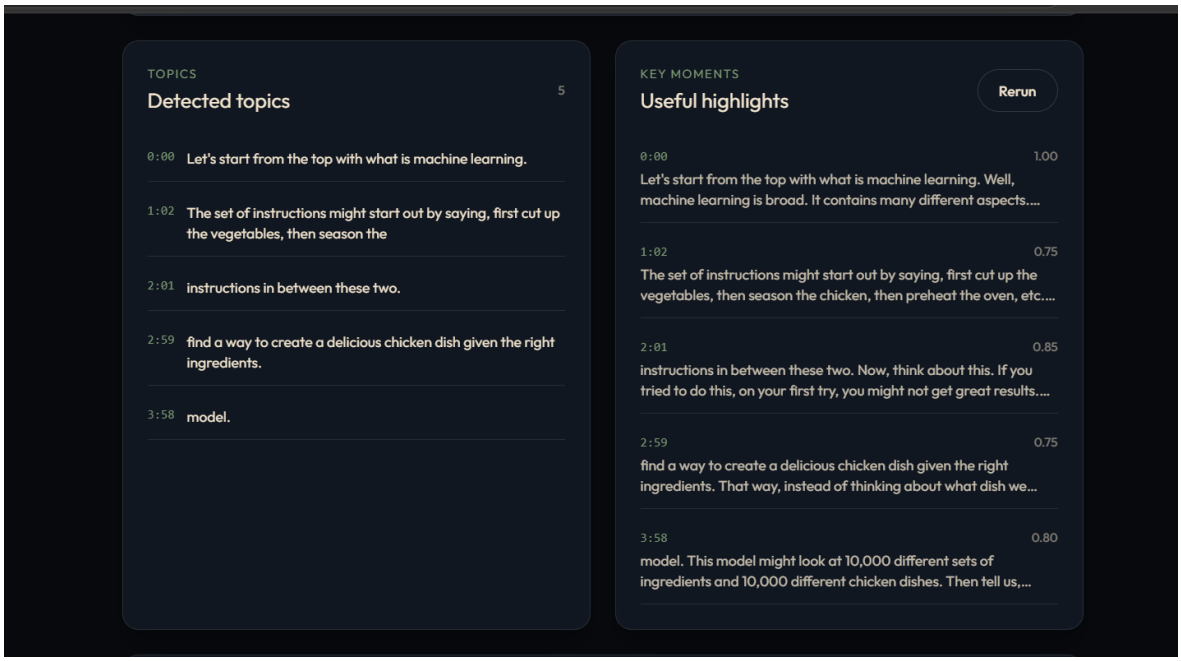
Outcome: Enables clear thematic segmentation of long-form videos, making transcripts navigable and searchable by topic.

2. Highlight Scoring

- **Scoring Formula (capped at 1.0)**:
    - 55% → Content density (information richness)
    - 30% → Presence of importance-marker keywords (important, definition, example, conclusion)
    - 15% → Positional signal (first/last sentence in a chunk)

  score=0.55(content density)+ 0.30(keyword signal)+0.15(position signal)

- **Candidate Selection**:
    - Ranked by score
    - Overlapping candidates suppressed
    - Up to 8 key moments persisted with:
        - Timestamp
        - Transcript text
        - Topic relation
        - Highlight type
        - Importance score

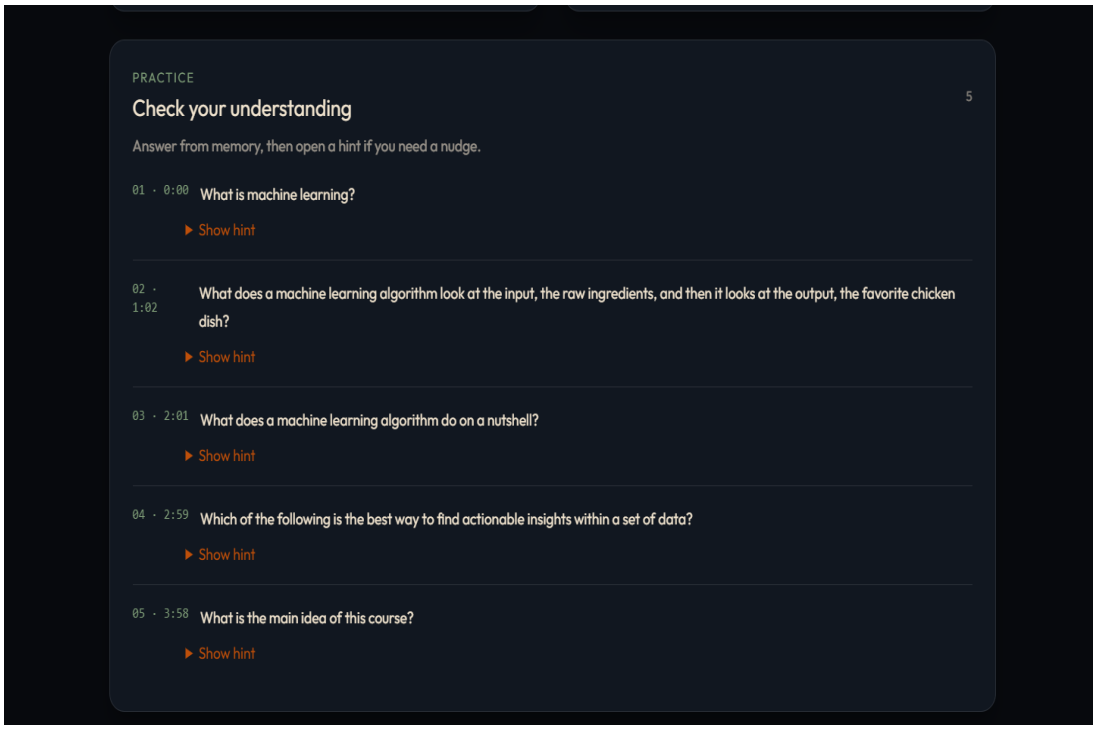


Outcome: Produces a concise timeline of high-value video segments for quick review.

3. Learning Questions

- **Selection**: High-scoring, non-overlapping chunks chosen.
- **Deterministic Generator**: Detects definition, process, cause, example, result, and general-claim language.

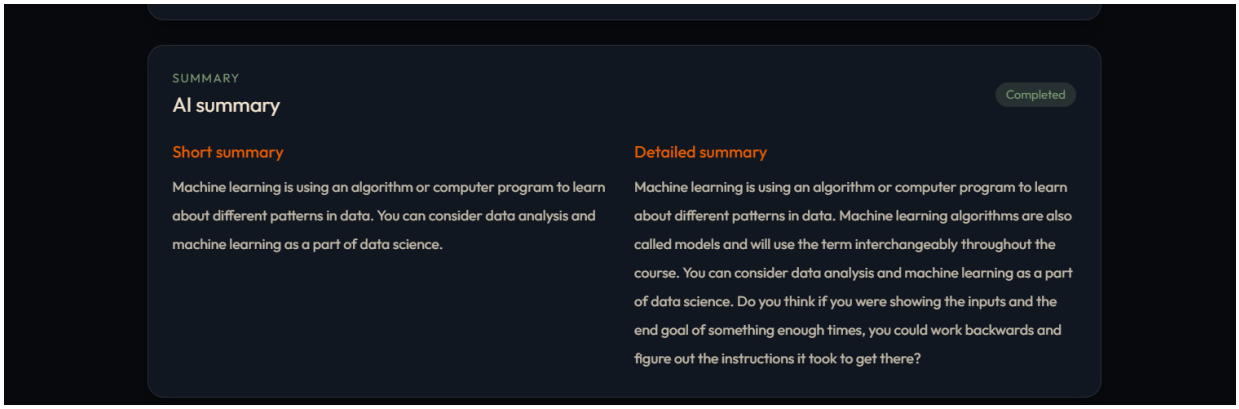
- **Optional AI Generator**:
    - Uses google/flan-t5-small to create a question + hint.
    - Fallback: deterministic prompts if generation fails.
- **Limit**: Up to 5 learning questions per video



Outcome: Converts passive video content into active learning material, supporting quizzes and comprehension checks.

4. Summaries

- **Detailed Summary**:
    - Transcript divided into 700-word chunks.
    - Each chunk summarized with facebook/bart-large-cnn.
    - Results joined into a comprehensive summary.
- **Short Summary**:
    - First 3,000 characters condensed into a shorter overview.



Outcome: Provides both granular and high-level summaries, enabling efficient content consumption.

5. Integrated Dashboard Features

- **Key Moments Timeline** – Interactive timeline with highlights and thumbnails.
- **Topic Map** – Visual segmentation of transcript topics.
- **Learning Questions Panel** – Auto-generated Q&A for learners.
- **Summary Viewer** – Switch between short and detailed summaries.
- **Engagement Analytics** – Track watch time, popular topics, and user activity.



6. Benefits

- 🎯 **Efficiency** – Learners and educators can jump directly to important segments.
- 📊 **Insightful Analytics** – Content creators gain visibility into engagement and topic popularity.
- 📚 **Active Learning** – Auto-generated questions transform videos into interactive study material.
- ⚡ **Scalability** – Modular design supports diverse video types (lectures, webinars, tutorials, media).